

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):	
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)	
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.	
ê by polarization plug into templates above	

Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{ } = (1 + \Gamma_{ }) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{ } ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_2}{\eta_1}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{ } ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{ } = 1 - R_{ }$

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; nonmag. ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η_i plug into table above	
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)}$	$\eta_0 = 120\pi \approx 377 \Omega$
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.	
Low-loss correction $\eta_c \sigma/(\omega\epsilon) \ll 1$	
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$	
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.	

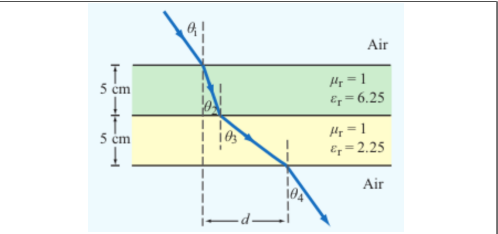
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	$\theta_{\text{exit}} = \theta_{\text{incident}}$
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
	$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$	
Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	$\eta = \eta_0 / \sqrt{\epsilon_r}$ exact $\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$
Low-loss	< 10 ⁻²	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon'}{2\omega \epsilon_r}\right)$ $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: $(\eta / \sqrt{X}) e^{j\theta_\eta}$ see below
Good cond.	> 10 ²	$\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params $\sigma/(\omega\epsilon) \ll 1, \mu_r = 1$	
$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), $\beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)	
Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$	
$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$	
$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$	

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$	TM mode $H_z = 0, E_z \neq 0$
Dimensions a × b with a > b along $\hat{x}, \hat{y}$; wave in $\hat{z}$.	
<i>E_x form (TE)</i> read m, n from arguments	
$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$	
<i>coef. on x = mπ/a; coef. on y = nπ/b.</i>	

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m=0 or n=0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode <i>generator: $\tilde{H}_z$</i>	$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TE}, \tilde{H}_y = \tilde{E}_x / Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	

TM mode <i>generator: $\tilde{E}_z$</i>	$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TM}, \tilde{H}_y = \tilde{E}_x / Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	

TEM plane wave *for reference*
 $\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x / \eta$

$$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega \sqrt{\mu\epsilon}$$

Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2}$ (Hz)	
$\beta = k \sqrt{1 - (f_c/f)^2}$ (rad/m)	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}}$ (m/s)	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) <i>result is unitless</i>	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)	
<i>Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10⁸ (m/s), m, n (unitless).</i>	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$ (m/s), $u_p u_g = u_{p0}^2$
$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower u _g ⇒ arrive later.

BOUNDARY

η _i , θ _t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = Γ ²
k _i = ω√ε _{r,i} /c (rad/m)
α ₂ (Np/m), β ₂ (rad/m): lossy
z = 0 — boundary plane (m)
η ₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ _i , θ _t — angles from normal
n _i = √ε _{r,i} — index of refr.
t _i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: E ⊥ this plane
: E this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f _{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z _{TE} /TM — wave imp. (Ω)
E _x — E-field comp. (V/m)
H _y — H-field comp. (A/m)
η = η ₀ /√ε _r

VELOCITIES

u _{p0} = c/√ε _r — bulk phase vel.
u _p = u _{p0} /√1 - (f _c /f) ²
u _g = u _{p0} √1 - (f _c /f) ²
u _p u _g = u _{p0} ²
t = L/u _g — travel time
c = 3 × 10 ⁸ m/s

POWER & UNITS

%P _r , %P _t — pwr refl./trans. (unitless)
%P _r = 100 Γ ²
%P _t = 100 τ ² η ₁ /η ₂
%P _r + %P _t = 100
σ (S/m), ε (F/m), μ (H/m)
ε ₀ = 8.85 × 10 ⁻¹² F/m
μ ₀ = 4π × 10 ⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 <i>l</i> = antenna rod length (m); <i>c</i> = 3 × 10 ⁸ m/s		
λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN <i>S</i> (<i>R</i> , θ), <i>D</i> = 1.5
= 1/4	Quarter-wave	ARB formula, π <i>l</i> /λ = π/4
= 1/2	Half-wave	half-wave shortcut, <i>D</i> = 1.64, <i>R</i> _{rad} = 73 Ω
= 1	Full-wave	ARB, <i>D</i> = 2.41, <i>R</i> _{rad} = 199 Ω
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — *S*(*R*, θ)

Far-field E,H phasors <i>small l</i> ≪ λ	
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$	$\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density <i>l</i> /λ < 1/50; <i>far field</i>	
<i>S</i> (<i>R</i> , θ) = <i>S</i> _{max} sin ² θ (W/m ²)	
<i>S</i> _{max} = $\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m ²)	
<i>l</i> : rod length (m)	<i>I</i> ₀ : peak current (A)
<i>R</i> : obs. distance (m)	θ: angle from dipole axis (rad)
<i>k</i> = 2π/λ (rad/m)	η ₀ = 120π ≈ 377 Ω
<i>S</i> _{max} : max pwr density (W/m ²)	
Max @ broadside ⊥ to dipole axis = equator plane θ _{max} = π/2; <i>D</i> exact <i>D</i> = 1.5 (1.76 dB)	
Total radiated power <i>P</i> _{rad} = $\frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — *S*(θ)

Power density at <i>R</i> any <i>l</i>	
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$	
At θ = 0, π: bracket is 0/0. <i>L'Hôpital</i> ⇒ <i>S</i> = 0 (no axial radiation). <i>TI-36X Pro</i> alt: evaluate bracket at θ = 0.001 <i>rad</i> → tiny → 0.	
Normalized pattern <i>dimensionless</i> <i>F</i> (θ) = <i>S</i> (θ)/ <i>S</i> _{max}	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion multiply by π/180 or 180/π						
θ _{rad} = θ _{deg} · $\frac{\pi}{180}$		θ _{deg} = θ _{rad} · $\frac{180}{\pi}$				
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	θ
0	0	0	1	0	1	
30	π/6	1/2	√3/2	1/4	3/4	
45	π/4	√2/2	√2/2	1/2	1/2	
60	π/3	√3/2	1/2	3/4	1/4	
90	π/2	1	0	1	0	
180	π	0	−1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

<i>N</i> identical elements along axis, spacing <i>d</i> , equal phase (broadside, θ _{max} = π/2). Total pattern <i>S</i> = <i>S</i> _{<i>e</i>} · <i>F</i> _{<i>a</i>} (<i>S</i> _{<i>e</i>} = single elem).	
Array factor (uniform, broadside) <i>k</i> = 2π/λ, γ = <i>k d</i> cos θ	
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\max} = N^2$ at θ = π/2	
Normalize <i>F</i> _{<i>a</i>} ^{norm} = <i>F</i> _{<i>a</i>} / <i>N</i> ²	
HPBW (broadside, uniform) use <i>N d</i> , not (<i>N</i> −1) <i>d</i>	
$\beta \approx 0.88 \frac{\lambda}{N d}$ (rad) = 50.8° · $\frac{\lambda}{N d}$	
Grating lobes appear if <i>d</i> > λ. Avoid.	

BEAM PARAMETERS — β, Ω_{*p*}, *D*

Normalize <i>always start here</i>		
$F(\theta) = S(\theta)/S_{\max}$ (unitless)		
Beamwidth β at threshold <i>X</i>		
Set <i>F</i> (θ _{<i>X</i>}) = <i>X</i> , solve for θ _{<i>X</i>} ; β = 2θ _{<i>X</i>} (symmetric)		
Threshold	<i>X</i> = 10dB/10	θ _{<i>X</i>} (Gauss. short-cut)
HPBW (−3dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any <i>F</i> (θ _{<i>X</i>})	any <i>X</i>	$F(\theta_X) = X$
Pattern solid angle Ω _{<i>p</i>} recognize pattern, look up		
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$ (sr)		
No φ dep.: Ω _{<i>p</i>} = 2π ∫ ₀ ^π <i>F</i> (θ) sin θ <i>d</i> θ.		
θ is the integration variable — DON'T plug in a number for θ (e.g. θ _{1/2}). Integrate over θ ∈ [0, π].		
Pattern <i>F</i> (θ)	Ω _{<i>p</i>} (sr)	
Isotropic (<i>F</i> = 1)	4π ≈ 12.57	
Hertzian sin ² θ	8π/3 ≈ 8.38	
Half-wave dipole	≈ 7.66	
Narrow Gauss <i>e</i> ^{−<i>a</i>θ²}	π/ <i>a</i> = πθ _{1/2} ² /ln 2	
2-axis HPBW (aperture)	β _{<i>xz</i>} · β _{<i>yz</i>}	
Other (use integral)	numerical	

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π.

Directivity <i>D</i> always <i>D</i> = 4π/Ω _{<i>p</i>} ; common shortcuts below	
$D = \frac{4\pi}{\Omega_p}$ (unitless)	<i>D</i> _{dB} = 10 log ₁₀ <i>D</i>
WORD TRAP: “direction” = θ _{max} (rad); “directivity” = <i>D</i> (unitless).	
Dipoles → COMMON DIPOLE PARAMS table. Aperture: <i>D</i> ≈ 4π <i>A_p</i> /λ ² . 2-axis HPBW: <i>D</i> ≈ 4π/(β _{<i>xz</i>} β _{<i>yz</i>}). Circular: <i>D</i> ≈ 4π/β ² .	
Gain (if asked) ξ = rad. eff., 0 < ξ ≤ 1	
<i>G</i> = ξ <i>D</i>	<i>G</i> _{dB} = 10 log ₁₀ <i>G</i>
Lossless / ideal antenna ⇒ <i>G</i> = <i>D</i> .	

COMMON DIPOLE PARAMETERS

Lossless (ξ = 1): <i>G</i> = <i>D</i> .				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< λ/50	1.5	1.76	80π ² (<i>l</i> /λ) ²
Half-wave	λ/2	1.64	2.15	73.2
Monopole	λ/4 (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: *P*_{rad} = 36.6 *I*₀² *W*. Monopole (λ/4 above ground): same patt. as half-wave but *P*_{rad}, *R*_{rad} halved.

ANTENNA AREAS (*A_e*, *A_p*)

Effective area antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above	
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter <i>d</i> ; <i>l</i> see table above	
<i>A_p</i> = <i>l d</i> (m ² , any dipole)	
Inverse: <i>D</i> from <i>A_e</i>	
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
Thin wire: <i>A_e</i> ≫ <i>A_p</i> (e.g., Hertzian <i>A_e</i> = 3λ ² /8π ≈ 0.119λ ²).	

FRIIS TRANSMISSION FORMULA

For each antenna's *G*: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (*G* = *D*, lossless). Else (dB/linear value given) → use as given (convert dB → linear).

Pwr density at RX, then RX pwr *G_t*, *G_r* linear gains; λ = *c*/*f*

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: *P_r* = *S_r* *A_{e,r}*.

Solve for *R* max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using *A_e* recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: *L_{fs}* = (4π*R*/λ)², so *P_r* = *G_t* *G_r* *P_t* / *L_{fs}*.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, *F_r* normalized patterns at the link directions; *F* = 1 when aligned.

Recipe 1. λ = *c*/*f*. 2. Convert any dB gain to linear: *G* = 10^{*G*_{dB}/10}.

3. Get *G_t*, *G_r* by antenna type:

- named dipole → COMMON DIPOLE PARAMS (*G* = *D* lossless)
- aperture / dish → APERTURE ANTENNAS (*G* = *D* ≈ 4π*A_p*/λ²)
- arbitrary pattern *F*(θ) → BEAM PARAMS (*G* = ξ · 4π/Ω_{*p*})
- given in dB → dB↔LINEAR table
- 4. Plug into Friis.

Use linear *G*, not dB. Convert FIRST.

dB ↔ LINEAR

Works for any power-like quantity (*G*, *D*, *P_r*/*P_t*, *SNR*, ...):

$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$		$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$	
dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For *E*-field / amplitude use 20 log not 10 log (power ∝ *E*²).

NOISE & SNR

Noise power thermal noise into matched receiver	
<i>P_N</i> = <i>k_B</i> <i>T</i> _{sys} <i>B</i> (W)	
<i>k_B</i> = 1.38 × 10 ^{−23} J/K; <i>T</i> _{sys} system noise temp (K); <i>B</i> receiver bandwidth (Hz).	
Signal-to-noise ratio	
<i>SNR</i> = $\frac{P_{\text{rec}}}{P_N}$ (unitless)	<i>SNR</i> _{dB} = 10 log ₁₀ $\frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs <i>SNR</i> > 10 dB for usable signal.	

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m²); *A_e* effective area (m², = λ² *D*/4π); β_{*xz*}, β_{*yz*} HPBWs in two planes (rad); *ℓ*, *d* aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths any aperture

$$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi / \beta^2$$

HPBW and <i>D</i> by aperture shape uniform illum.; β in rad			
Shape	<i>A_p</i>	HPBW (rad)	<i>D</i> (unitless)
Rect. <i>ℓ_x</i> × <i>ℓ_x</i> <i>ℓ_y</i>		β _{<i>x</i>} ≈ 0.88λ/ <i>ℓ_x</i>	<i>D</i> ≈ 4π/(β _{<i>x</i>} β _{<i>y</i>})
		β _{<i>y</i>} ≈ 0.88λ/ <i>ℓ_y</i>	= 4πℓ _{<i>x</i>} ℓ _{<i>y</i>} /λ ²
Square <i>ℓ</i>	<i>ℓ</i> ²	β ≈ 0.88λ/ <i>ℓ</i>	<i>D</i> ≈ 4π/β ² = 4πℓ ² /λ ²
Circular (diam. <i>d</i>)	π <i>d</i> ² /4	β ≈ 1.02λ/ <i>d</i>	<i>D</i> ≈ 4π/β ² = π <i>d</i> ² /λ ²

Scaling (any ratio) *D* ∝ *A_p*/λ²; β ∝ λ/√*A_p*

$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area → *D* × 2, β × 1/√2; double freq → *D* × 4, β × 1/2.

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
 a — coef. on θ^2 in Gaussian
 β — HPBW (rad or °)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ rad. eff.; $G = \xi D$
 $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
 P_r — power received (W)
 G_t — transmit gain (linear)
 G_r — receive gain (linear)
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\max} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)